

3. Auslander, D. M. and Kempf, C. J., *Mechatronics: Mechanical System Interfacing*, Prentice Hall Inc., New Jersey (1996).

4. John W. Webb & Ronald A. Reis, *Programmable Logic Controllers – Principles and Applications*, Fifth Edition, Pearson Education (2008).

5. *Workbook of Pneumatic, Electropneumatics and PLC* published by FESTO.

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessional (Assignments/Projects/Quizzes/Lab Evaluations/Reports, etc.)	30

UME420 : Operations Management

L	T	P	Cr
2	2	0	3.0

Course Objective: The objective of this course is to develop understanding of the strategic role of operations management in creating and enhancing a firm's competitive advantages.

Approved in 113th meeting of the Senate held on September 7, 2024.

This will help to apply key concepts and issues of operations management in both manufacturing and service organizations by enabling the students to apply analytical skills and problem-solving tools for the analysis of the operations problems like forecast demand, material requirement planning, inventory etc.

Syllabus

General: Operations Management: meaning and scope; significance of operations management in increasing productivity of firms; soft variety and hard variety; categories of production systems and layouts.

Forecasting Analysis: Need and benefits, various qualitative and quantitative models, error analysis in quantitative forecasting.

Production Planning: Aggregate production planning, pure and mixed aggregate planning strategies; Master production scheduling; material requirements planning and manufacturing resource planning (MRP I and MRP II); Supply Chain Management.

Inventory Management and Control: Need and types inventory, methods of handling inventory uncertainties, methods of inventory control systems, perpetual (fixed order-quantity) system, periodic (fixed order-interval) system, economic run length

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze the fundamental theory of operations and production management.
2. Analyze forecasting problems or issues faced by service and manufacturing industries.
3. Solve problems on materials requirement planning, aggregate production planning.
4. Analyze inventory management problems.

Text Books

- 1 Monks, J. G., Operations Management: Theory and Problems, McGraw Hill, New York (1987).
- 2 Krajewski, L. J., Ritzman, L. P. and Malhotra, M. K., Operations Management, Prentice Hall, New Delhi (2009).